



American International University-Bangladesh (AIUB)

Department of Computer Science

Faculty of Science & Technology (FST)

Software Engineering

**AIUB Portal Redefined: Integrating Campus Life with
Smart Digital Innovation**

Submitted To

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Project Proposal

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PROJECT PROPOSAL

Project Title

AIUB Portal Redefined: Integrating Campus Life with Smart Digital Innovation

1 Background to the Problem

1.1 Background Description

The American International University-Bangladesh (AIUB) Portal (<https://portal.aiub.edu/>) serves as the primary digital interface for students, faculty, and administrators. However, inefficiencies in course registration, campus navigation, resource access, communication, printing, complaint handling, gymnasium services, file sharing, and class scheduling lead to delays, user frustration, reduced engagement, and administrative bottlenecks. These issues undermine AIUB's vision of a technologically advanced, student-centric campus, necessitating a comprehensive digital overhaul.

1.2 Root Cause of the Problem

1. **Course Registration:** Manual processes and lack of demand-based allocation cause course conflicts and long queues.
2. **Lost and Found:** No centralized digital system results in untracked items and inefficient manual reporting.
3. **Campus Navigation:** Absence of precise indoor navigation leads to confusion in AIUB's multi-building campus.
4. **Event Notices:** Fragmented communication channels cause missed opportunities for seminars, exams, and club activities.
5. **Gymnasium Registration:** Offline registration and payment processes are time consuming and error prone.
6. **Complaint System:** Lack of security, anonymous reporting discourages students from addressing harassment or issues.
7. **Course Resources:** Limited digital access to reference books and materials restricts academic support.
8. **Printing System:** Dependency on local PCs and offline payments limits accessibility and convenience.
9. **Faculty Information:** Scattered or outdated faculty details hinder communication and coordination.
10. **File Sharing:** Reliance on external tools (e.g., Teams, WhatsApp) lacks security and campus integration.
11. **Class Scheduling:** Manual routine creation leads to conflicts and inefficient course planning.

1.3 Importance of Solving This Problem

1. **Course Registration:** Streamlining ensures equitable course access, reduces administrative workload, and enhances academic fairness.
2. **Lost and Found:** A digital system improves item recovery rates and fosters trust in campus services.
3. **Campus Navigation:** Precise navigation saves time, improves accessibility for new or disabled students, and enhances campus efficiency.
4. **Event Notices:** Centralized notices boost participation in academic and extracurricular activities, strengthening community engagement.
5. **Gymnasium Registration:** Online processes promote student wellness and reduce administrative errors.
6. **Complaint System:** Anonymous reporting enhances campus safety and inclusivity, addressing legal and cultural needs.
7. **Course Resources:** Digital access supports academic success and reduces financial burdens on students.
8. **Printing System:** Cloud-based printing improves accessibility and operational efficiency across the campus.
9. **Faculty Information:** A centralized directory enhances communication and academic planning.
10. **File Sharing:** Secure, integrated sharing improves privacy and reduces reliance on external tools.
11. **Class Scheduling:** Automated scheduling minimizes conflicts, optimizing time for students and faculty.

2 Solution to the Problem

2.1 Project Objective

To transform the AIUB Portal into a comprehensive, smart digital platform leveraging AI, AR, and cloud technologies to address inefficiencies in course registration, lost and found, campus navigation, event notices, gymnasium services, complaints, course resources, printing, faculty information, file sharing, and scheduling, thereby enhancing user experience and operational efficiency.

2.2 Solutions Provided to Solve the Problems

1. CGPA & Demand-Based Course Registration:

- Automate registration with CGPA-based slots (3.50–4.00, 3.00–3.49, <3.00) and batch-wise time windows.
- Analyze pre-midterm forms (CGPA, credits, course preferences) to open sections.
- Implement waitlisting and advisor-approved priority requests.

2. **Lost and Found Submission:**
 - Allow users to submit reports with descriptions and images, stored in a searchable database.
 - Require admin approval for public visibility; support updates/removals.
3. **Smart Location System (AI + AR Navigation):**
 - Provide real-time campus maps with AR overlays, BLE beacons, and AI image recognition.
 - Offer turn-by-turn guidance, voice prompts, and accessibility modes for wheelchair-friendly routes.
4. **Event and Notice Display:**
 - Display categorized notices (previous, current, upcoming) with filtering and event registration.
 - Restrict publishing to admins.
5. **Gymnasium Registration & Payment:**
 - Enable online registration and payment tracking with admin verification.
 - Provide membership history and renewal tracking.
6. **Anonymous Complaint System:**
 - Support encrypted, anonymous submissions accessible only to higher authorities.
 - Allow status tracking without identity exposure.
7. **Course Book Reference & Online Resources:**
 - Maintain a section for course books and online resources, searchable and downloadable.
8. **Global Printing System:**
 - Allow remote document uploads, cloud-based printing, and secure PIN- based release.
 - Integrate online payments (bKash, bank cards) with usage reports.
9. **Faculty Information Directory:**
 - Provide searchable faculty profiles with department, contact, courses, and research details.
 - Admin-managed updates.
10. **AIUB File Sharing System:**
 - Enable secure file sharing with temporary links, auto-deleting after expiry.
11. **Class Routine & Course Generator:**
 - Detect schedule conflicts and suggest eligible courses based on prerequisites.
 - Generate personalized routines.

2.3 Why This Solution is Particularly Appropriate

1. **Course Registration:** AI-driven demand analysis and CGPA prioritization ensure equitable access, reducing conflicts and manual advising.
2. **Lost and Found:** A searchable database with admin oversight streamlines item recovery, addressing manual process inefficiencies.
3. **Smart Location System:** AR and BLE beacons provide precise indoor navigation, overcoming GPS limitations in multi-building campuses.
4. **Event Notices:** Centralized, filterable notices with registration enhance engagement, solving fragmented communication.
5. **Gymnasium Registration:** Online processes with secure payments improve convenience, addressing offline bottlenecks.
6. **Complaint System:** Encrypted, anonymous submissions promote safety and inclusivity, encouraging issue reporting.
7. **Course Resources:** Digital access reduces costs and enhances academic support, solving limited availability.
8. **Printing System:** Cloud-based printing with online payments eliminates local PC dependency, enhancing accessibility.
9. **Faculty Information:** A centralized directory improves communication, addressing scattered data issues.
10. **File Sharing:** Secure, integrated sharing enhances privacy and efficiency, reducing external tool reliance.
11. **Class Scheduling:** Automated conflict detection optimizes planning, minimizing manual errors.

2.4 Feasibility to Meet Business Objective

- **Technical Feasibility:** Uses proven technologies (React, Node.js, ARKit, AR- Core, AWS) for scalability and performance.
- **Operational Feasibility:** Modular design and user-friendly interfaces ensure adoption by students, faculty, and admins.
- **Financial Feasibility:** Cloud infrastructure and modular development align with AIUB's budget; premium features enable monetization.
- **Business Alignment:** Supports AIUB's mission of digital transformation, enhancing student satisfaction and operational efficiency.

2.5 Basic Functionalities and Technological Impact

1. **Course Registration:** AI-driven allocation ensures fairness, reducing advising queues (societal impact: equitable education).
2. **Lost and Found:** Searchable database improves recovery rates (societal impact: trust in campus services).
3. **Smart Location System:** AR overlays and BLE beacons provide precise navigation with accessibility modes (societal impact: inclusivity).

4. **Event Notices:** Filterable bulletin with registration boosts participation (cultural impact: community engagement).
5. **Gymnasium Registration:** Online enrollment promotes wellness (health impact: student fitness).
6. **Complaint System:** Encrypted submissions encourage reporting (safety/legal impact: campus safety).
7. **Course Resources:** Digital access supports learning (societal impact: academic equity).
8. **Printing System:** Cloud-based printing enhances convenience (societal impact: operational efficiency).
9. **Faculty Directory:** Searchable profiles improve coordination (cultural impact: collaboration).
10. **File Sharing:** Secure links ensure privacy (legal impact: data protection).
11. **Class Scheduling:** Automated routines optimize time (societal impact: efficiency).

2.6 Target Users and Benefits

- I. **Students:** Streamlined registration, navigation, and resource access save time and enhance academic success.
- II. **Faculty:** Centralized directory and notices improve communication and teaching efficiency.
- III. **Administrators:** Automated systems and analytics reduce workload and improve decision-making.
- IV. **Disabled Users:** Accessibility modes in navigation ensure inclusivity.
- V. **Overall Impact:** Enhanced experience, safety, and engagement foster a connected campus community.

2.7 Scientific Contribution

- I. Advances AI-driven educational systems through demand-based registration and analytics.
- II. Contributes to indoor navigation research with AR and BLE-based solutions for campuses.
- III. Provides a scalable, documented model for university portal modernization, validated by metrics and feedback
- IV. Promotes interdisciplinary research in AI, AR, and cloud computing in education.

2.8 Literature Review

• Existing Studies:

- I. Khan et al. (2020): AI-based scheduling lacks demand-based allocation. Our system adds AI-driven demand analysis.
- II. Smith & Johnson (2021): AR navigation uses GPS, ineffective indoors. Our BLE and QR code-based AR enhances precision.

- III. Lee et al. (2022): Cloud printing for enterprises lacks campus customization. Our SafeQ-like system integrates payments.
- IV. Rahman (2023): Anonymous complaint systems lack encryption details. Our solution adds security and tracking.

- **Extension of Studies:**

- I. Integrates AI, AR, and cloud computing for a holistic campus solution.
- II. Introduces demand-based registration for equitable access.
- III. Enhances indoor navigation with AR and BLE.
- IV. Adds engagement features (event registration, file sharing).

2.9 Existing Software Solutions

- I. **Blackboard/Canvas:** Course management but lacks AR navigation, demand-based registration, or complaints.
- II. **Google Maps Indoor:** Basic indoor mapping without AR overlays or university specific features.
- III. **SafeQ /Papercut:** Cloud printing lacks campus-specific payment integration.
- IV. **Microsoft Teams:** File sharing without campus integration or temporary links.

2.10 Extension of Existing Solutions

- I. **Customization:** Integrates all services into an AIUB-specific platform.
- II. **Advanced Technology:** Combines AI, AR, and cloud computing for a seamless experience.
- III. **Enhanced Benefits:** Offers precise navigation, equitable registration, secure complaints, and accessible resources.
- IV. **Scalability and Inclusivity:** Cloud architecture supports growth; accessibility modes ensure inclusivity.

Conclusion

The "**AIUB Portal Redefined: Integrating Campus Life with Smart Digital Innovation**" project proposes a transformative solution to address critical inefficiencies in the AIUB Portal. By leveraging AI, AR, and cloud technologies, the enhanced platform will streamline course registration, campus navigation, event notices, gymnasium services, anonymous complaints, course resources, printing, faculty information, file sharing, and class scheduling. These solutions tackle root causes such as manual processes, fragmented services, lack of accessibility, delivering equitable education, improved safety, and enhanced user engagement. The project's modular design, advanced technology stack, and Agile methodology ensure feasibility, scalability, and alignment with AIUB's mission of digital transformation. By extending existing research and software solutions, this initiative offers a scalable, inclusive model for modernizing university portals, fostering a connected, efficient, and student-centric campus community.

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